

Indic NLP · Bhashini · AI4Bharat

English-only apps max out at 150 million. Indic-first apps reach 600 million. Whoever solves Indic best wins the next half-billion users. That is the Indian AI moat.

Which languages does your current product support? Which are missing?

WHAT'S INSIDE

- 01** The data gap
- 02** Bhashini — the national platform
- 03** AI4Bharat — the research engine
- 04** Hinglish classifier — 25 lines
- 05** Voice-first design

WHY THIS LESSON MATTERS

62% of Indian internet users prefer native-language content.

NUMBER 1

150 million

From real-world deployments.

NUMBER 2

600 million

From real-world deployments.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Why Indic NLP is underserved.

2

Use Bhashini APIs.

3

AI4Bharat models.

4

Hinglish classifier with IndicBERT.

5

Voice-first Indic product.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

The data gap

Why frontier LLMs fall short on Indic

02

Bhashini — the national...

Free government APIs for Indic AI

03

AI4Bharat — the research...

IIT Madras initiative + Nilekani
Philanthropies. Open weights + free
use.

04

Hinglish classifier — 25...

IndicBERTv2 embeddings

01

PART 1 OF 4

The data gap

Why frontier LLMs fall short on Indic

CONCEPT 1 OF 4

The data gap

English is 50% of the internet.

Indic langs: 15% of humanity but < 2% of training data.

Hindi passably handled; Tamil poorly; Kannada laughably.

Gap = pain point for users AND opportunity for builders.

MOAT

Indic first

Few frontier labs will prioritise this — Indian teams must.

DEFINITION

The data gap

Why frontier LLMs fall short on Indic

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

Bhashini — the national platform

Ministry of Electronics and IT, launched 2022.

22 scheduled languages.

Speech, text, translation, OCR.

REST APIs; register at bhashini.gov.in.

Public-sector deployments mandate Bhashini integration.

FREE

bhashini.gov.in

Get keys; go.

KEY POINTS

- Ministry of Electronics and IT, launched 2022
- 22 scheduled languages
- Speech, text, translation, OCR
- REST APIs; register at bhashini.gov.in

AI4Bharat — the research engine

IIT Madras initiative + Nilekani Philanthropies. Open weights + free use.

DEFINITION

AI4Bharat — the research engine

IIT Madras initiative + Nilekani Philanthropies. Open weights + free use.

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Hinglish classifier — 25 lines

Tokenise mixed-language text.

Get 768-dim embeddings.

Train a small classifier on top.

Handles Roman + Devanagari in the same input.

USE

IndicBERTv2

Open-weight; accepts code-mix.

KEY POINTS

- Tokenise mixed-language text
- Get 768-dim embeddings
- Train a small classifier on top
- Handles Roman + Devanagari in the same input

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Frontier LLM vs AI4Bharat on Tamil

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Which Indic language would you ship first — and why?

Q2

What does your grandmother's dream app look like?

Q3

Where does code-switching break current products you use?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

Why Indic NLP underserved:

A

Easy

B

Low data; low priority at frontier labs

C

Illegal

D

Nothing

Why Indic NLP underserved:

CORRECT ANSWER



Low data; low priority at frontier labs

WHY

< 2% of training data; frontier labs prioritise English

Bhashini ministry:

A

Finance

B

Education

C

MeitY

D

Health

Bhashini ministry:



CORRECT ANSWER

MeitY

WHY

Ministry of Electronics and IT

AI4Bharat host institution:

A

IIT Bombay

B

IIT Madras

C

IISc

D

ISI

AI4Bharat host institution:



CORRECT ANSWER

IIT Madras

WHY

IIT Madras

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

English is 50% of the internet.

Indic langs: 15% of humanity but < 2% of training data.

Hindi passably handled; Tamil poorly; Kannada laughably.

Gap = pain point for users AND opportunity for builders.

MOAT

Indic first

Few frontier labs will prioritise this — Indian teams must.

THE TAKEAWAY

"The next 500 million users do exactly that. Build for them first."

WHAT TO NOTICE

1

The data gap

2

Bhashini — the national...

3

AI4Bharat — the research...

4

Hinglish classifier — 25...

Hinglish sentiment

500 tweets.

DELIVERABLES

- IndicBERTv2 embeddings.
- Logistic regression head.
- Report macro-F1.
- Document one decision you made because of this lesson.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Macro-F1 on forum.

ONE THING TO REMEMBER

“

"The next 500 million users do exactly that. Build for them first."

— AI Learning Hub

END OF LESSON 7

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → **Projects Saturday.**